

Multiple Choice (10 marks)

1. Find $v(0^+)$ and $(dv/dt)_{t=0^+}$ if $V(s) = [(-36s^2 - 24s + 2) / (12s^3 + 17s^2 + 6s)]$
 - (a) $v(0^+) = -4, (dv/dt)_{t=0^+} = 2.25$
 - (a) $v(0^+) = -2, (dv/dt)_{t=0^+} = 3.25$
 - (a) $v(0^+) = -3, (dv/dt)_{t=0^+} = 4.25$
 - (a) $v(0^+) = 0, (dv/dt)_{t=0^+} = 1.5$
 - (a) $v(0^+) = -3, (dv/dt)_{t=0^+} = 1.25$
2. Find the energy stored in a nonlinear capacitor with characteristics given by $g(t) = 0.5 \nu^2(t)$ and $\nu(t) = 1 + 0.5 \sin t$ from $t = 0$ to $t = 1$ sec.
 3. 0.785 J
 4. 0.211 J
 5. 0.125 J
 6. 0.623 J
 7. 0.956 J

An equation of state due to Dieterici is $P(V - nb') \exp [na' / (RTV)] = nRT$. Evaluate the constants a' and b' in terms of the critical constants P_c, V_c , and T_c of a gas.

- (a) $a' = 2 RT_c V_c, b' = (2 * V_c)$
- (b) $a' = P_c V_c^2, b' = V_c / 3$
- (c) $a' = RT_c^2 / P_c, b' = V_c / P_c$
- (d) $a' = 2 RT_c V_c, b' = (V_c / 2)$
- (e) $a' = 2 RT_c, b' = V_c$

Magnetic flux has the unit of

- (a) Volt
- (b) Weber
- (c) Watt
- (d) Ohm
- (e) Joule

Find the theoretical variance of the random variable with the following probability distribution.

x	Pr(X = x)
0	1/4
1	1/2
2	1/8
3	1/8

- (a) 0.625
- (b) 0.683
- (c) 0.350
- (d) 1.125
- (e) 1.000

True / False (10 marks)

Answer True or False

6. True or False: The inverse Laplace transform of $F(s) = [(3s + 1) / \{(s - 1)(s^2 + 1)\}]$ is $e^t - \cos t + \sin t$.
7. True or False: The correct answer to 'Consider a sphere of radius a containing charge of constant density $\rho = \text{const}$, inside, while $\rho = 0$ outside. Find ...' is ' $\phi(r) = [(3\rho a^3) / (3\epsilon_0 r)]$, $\phi(r) = [\rho a^2 / (6\epsilon_0)] (3a^2 - r^2)$ '.
8. True or False: In toggle mode, a JK flip-flop can be configured with $J = 0$ and $K = 0$, resulting in a state change with every clock pulse.
9. True or False: The equation $y[n] = y[n-1] + 2n - 1$ correctly represents the sum of squares of integers from 0 to n .
10. True or False: The z-transform of the step-sequence $f[n] = u[n]$ is given by $F(z) = 1 / (z - 1)$ for $|z| > 1$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the heat transfer process between a spherical automobile component with a diameter of 6 inches at 80°C and a cubical oven at 560°C . Calculate the heat transferred, considering both convection and radiation.
11. Discuss the isothermal compression of a liquid in a chamber, deriving the expression for the total work done during a quasistatic process described by $\ln(V/V_0) = -A(p - p_0)$.

End of Paper